Table 1: Revision History

Date	**Developer(s)**	**Change**
2026-04-05	Jeremy Orr	Initial user guide
2026-04-05	—	Restructured around user workflows; matches app navigation order
2026-04-05	—	Added per-area overviews (Home through About) before workflows

Maintainers: add a row when user-visible behaviour changes.

User Guide
Software Engineering (InfraRAT Web Application)

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1 How to use this guide

This guide is written for **you**—someone using InfraRAT in a web browser. You do not need technical setup.

- Read Section 5 for a short menu map, then Section 7 if you want a fuller picture of what each screen contains.
- Skip to any **workflow** in Section 8 that matches what you want to do today. Each workflow is a numbered checklist you can follow in order.
- Use Section 9 for file types, short definitions, and fixes when something blocks you.

2 Introduction

The web application is branded **InfraRAT**. It supports data analysis for animal models of OCD on a large rat behavioural dataset (on the order of twenty thousand trials): natural-language questioning, tabular query results, inventory counts and session lists, charts and heatmaps, pre-built comparison dashboards, and per-session trajectory inspection.

This document explains what you see in the app and how to complete common tasks step by step. The product name in the header and footer is InfraRAT; Software Engineering is the capstone project name on the title page.

3 Quick start

1. Open the web address your administrator gave you.
2. Read **Home** for an overview, carousel cards, and example query wording.
3. On **Query**, use **Ask** for conversational answers, or **Select** for tables and (when applicable) downloads.
4. Open **Inventory** (*Data Inventory*) to count sessions and files, optionally list up to 500 sessions, and download bundles.
5. Use **Visualizations** for bar chart, heatmap, spatial heatmap, velocity profile, or **Query Visualizations**.
6. Use **ToolBox** to load one session, inspect tables and the path view, and read summary metrics.
7. Open **About** (info icon in the header) for team information and FAQs.

4 What is InfraRAT?

InfraRAT helps you explore a large rat behavioural dataset (tens of thousands of trials). In plain terms you can: ask questions, filter and count sessions, open charts and maps, compare groups on standard measures, inspect one session closely, and download selected file types for the sessions you care about.

The course project name is Software Engineering; **InfraRAT** is the name in the app header.

5 How the site is organized

The top bar matches the order below (on a phone, open the menu icon first). This is the same order as the workflows in the next section.

Home
: Orientation: what the platform does, rotating cards, and example wording (start here on day one).

Query
: Ask questions in chat, or switch to Select to get tables and optional downloads (*Talk to the FRDR*).

Visualizations
: Pick a tool from the gallery (bar chart, heatmap, maps, velocity, dashboards).

Inventory
: *Data Inventory*—counts and filters across the dataset; list sessions and download bundles.

ToolBox
: *Analysis Toolbox*—one session at a time: tables, path view, summary numbers.

To the right: **About** (info icon) for team and FAQs; **GitHub** for reporting issues. The footer links to FRDR and the lab collection.

Tip: many tasks use two areas—for example Inventory to find IDs, then ToolBox or Visualizations to go deeper.

6 Access and environment

There is no login or single sign-on screen in the current app: you use the deployment URL you were given.

If the header shows a **STAGING** badge next to the product name (for example when the app is run on a staging port), treat that environment as non-final.

7 Overview of each area

The subsections below describe *what each part of the app is for* and what you will see there. Step-by-step instructions are in Section 8.

7.1 Header, footer, and navigation

The header stays visible as you scroll. It shows the InfraRAT logo and name. The main destinations are **Home**, **Query**, **Visualizations**, **Inventory**, and **ToolBox**. On a narrow screen, open the menu control to reach the same links.

On the right, the **About** control (info icon) opens the About page. The **GitHub** icon opens the project issue tracker in a new tab (for feedback or bug reports).

The footer shows the app version (e.g. InfraRAT v1.0), course attribution, and links to FRDR and the Szechtman lab collection. If you open a bad link inside the app, you may see a simple "not found" page.

7.2 Home

Home introduces the platform: purpose, scale (twenty-thousand-plus trials), and the idea of going from raw trials to insights.

An auto-advancing **carousel** presents use cases—dataset access and download, visualization and analysis, natural language questions, experiment playback, processing pipeline, and API-related messaging. These are overview cards, not separate apps.

Lower on the page, images contrast video trials with temporal or tabular views. Example natural-language prompts appear in quotation marks. A **sample results table** (trial ID, group, checking count, home-base duration, video sync, etc.) shows the *shape* of tabular output; it is demo content, not live results from your run.

7.3 Query

Query is the main chat-style interface to the dataset. The page title is *Query*; the subtitle is *Talk to the FRDR* (FRDR is the Federated Research Data Repository, also referenced when you download bundled files).

A switch toggles **Ask** and **Select**. In **Ask**, you type in everyday language and read a streamed answer in chat bubbles; emphasized phrases may appear in bold. In **Select**, the same kind of input produces a structured reply: *Query Plan*, *SQL Query*, and *Results*. The embedded results table shows up to fifty rows, with a note if there are more.

Your messages appear on the right; the system's replies appear on the left.

Use the bottom text area (placeholder *Ask, Search or Chat...*). Press **Enter** to send and **Shift+Enter** for a new line without sending.

After a successful **Select** answer, a **Sessions** section below lists the full result set in a wide, scrollable table.

Download Session Files appears on this page only when the last message in the thread is a **Select** result (not a plain **Ask** reply). The dialog offers CSV, EWB, GIF, JPG, and MPG; after you confirm, a progress bar runs and your browser saves a ZIP whose name begins with `FRDR_FILES_`.

7.4 Data Inventory

Inventory opens the *Data Inventory* page. It explains that counts are by raw data type (video, track files, and related types) and that filters narrow which sessions are included.

Show full inventory and **Unoperated rats only** are shortcuts at the top. The left sidebar holds expandable filter groups: drug treatment (you may check several), file types, fine-grained drug regimens (labelled *Drug Regiments* in the app), brain manipulation (surgery type and region), apparatus (including pattern and room), and session type. **Apply** runs the search; **Clear** resets choices and clears results and the session table.

After counts load, the main area shows how many sessions match, per-type file and session counts (with info tooltips), and a **Counts by data type** bar chart. **View sessions (up to 500)** loads a scrollable session table. **Download Session Files** in the sidebar is for the filtered list: it becomes available only after you have loaded that table and it is not empty. Format choices match the Query download dialog.

7.5 Visualizations

Visualizations opens a **gallery** of tiles. Each tile has a short description and an **Open** button. The gallery includes **Bar Chart**, **Heatmap**, **Spatial Heatmap**, **Velocity Profile**, and **Query Visualizations**. There is no tile for **Line Chart** on the gallery; that builder still exists in the app—ask your administrator for a bookmark or link if you need trends over time.

Bar chart builder: choose **X-Axis (Group By)** and **Y-Axis**; axis titles and units come from the data service.

Line chart builder (when you have a link): choose **X-Axis (Time Period)** and a **Y-Axis** metric for trends over time.

Heatmap builder: choose two dimensions and a **metric** for cell colour.

Spatial heatmap: combine one or more sessions, bin size, visits vs time in bin, optional fixed colour cap, optional **home base** rectangle, hover details on the map, and export to CSV or PNG.

Velocity profile: pick a session, define a circular zone (centre and radius), set frame windows and minimum trip length, choose colouring mode, then **Generate** to see entering and exiting speed traces.

Query Visualizations: choose a **Graph Set** (e.g. Tucci-style snapshot bars or Ballester-style lines across injections). Panels cover measures such as checking frequency, check length, recurrence time, stops before returning, and rest duration. Long on-page text describes the comparison design. Partial failures show a warning while other panels may still load.

7.6 Analysis ToolBox

ToolBox shows the *Analysis Toolbox* page. Everything here is about **one session at a time**.

You search or pick a session ID, then **Load Session**. On success you typically see **Session Info** (metadata), **Data Points** (trajectory or frame table), a still **image** when available, and an **interactive path** plot of movement.

Three cards summarize **Distance Travelled** (metres), **Total Number of Checks**, and **Average Check Time** (shown as $\log_{10}$ seconds). If distance is not available, **Generate Distance?** tries to compute it from identifiers in session info.

If the session cannot be loaded, the app states that it is not supported.

7.7 About

About lists faculty and student contributors and credits a prior capstone team. An accordion holds FAQs (purpose, who can use the platform, what data exists, downloads). A button can take you to **Query** if you want to ask a follow-up question.

7.8 Downloads and file formats (summary)

ZIP bundles from **Query** (after **Select**) and **Inventory** (after loading the session list) can include CSV, EWB, JPG, MPG, and GIF as you select them. Spatial heatmap exports use separate CSV/PNG actions on that page. Full detail is in Section 9.1.

8 Guided workflows

Each workflow below is meant to be followed top to bottom. Screen names and button labels match the app.

8.1 Workflow A: First visit—orient yourself

1. Open **Home**. Read the headline and short description of scale.
2. Watch the **carousel**; each card states a capability (access, visualization, natural language, and so on).
3. Skim the example questions and the **sample table**. The table is illustrative only—it shows the *shape* of results, not live data.
4. When you are ready to try the app, go to Workflow B or C.

8.2 Workflow B: Ask a question (no tables or downloads)

1. Click **Query** in the top bar.
2. Leave the switch on **Ask** (left side).

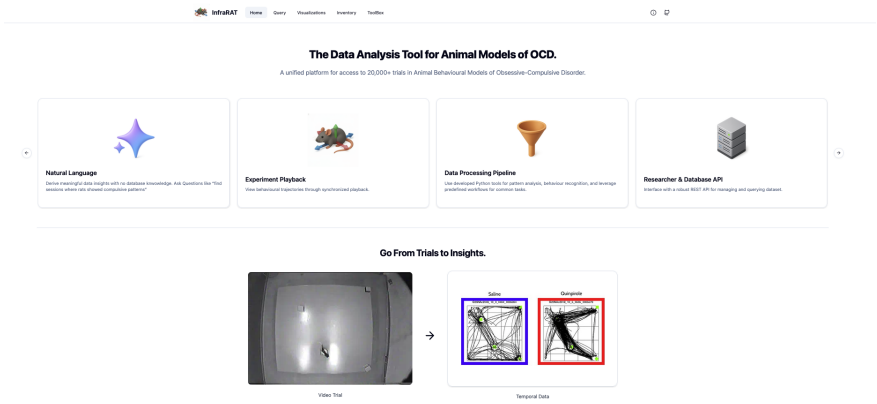

Figure 1: Home: start here to understand what InfraRAT offers.

3. Type in the box at the bottom ("Ask, Search or Chat…"). Press **Enter** to send, or **Shift+Enter** for a new line.

4. Read the answer as it appears on the left. Important phrases may appear in **bold**.

Use this when you want an explanation or exploration, not a spreadsheet or ZIP file.

8.3 Workflow C: Get a table of sessions from a question (and optionally download)

1. Go to **Query**. Flip the switch to **Select**.

2. Type what you want in plain language; press **Enter** (or use the send button).

3. On the left, read the card: *Query Plan*, then *SQL Query*, then *Results*. The small table shows up to fifty rows; a note appears if there are more.

4. Scroll down to **Sessions** to see the full table for that answer (wide columns—scroll sideways if needed).

5. To download files for those sessions: click **Download Session Files**. It is enabled only if the **last** message in the thread is a Select result (not Ask). Choose CSV, EWB, GIF, JPG, and/or MPG, confirm, wait for the progress bar, then save the ZIP (name begins with `FRDR_FILES_`).

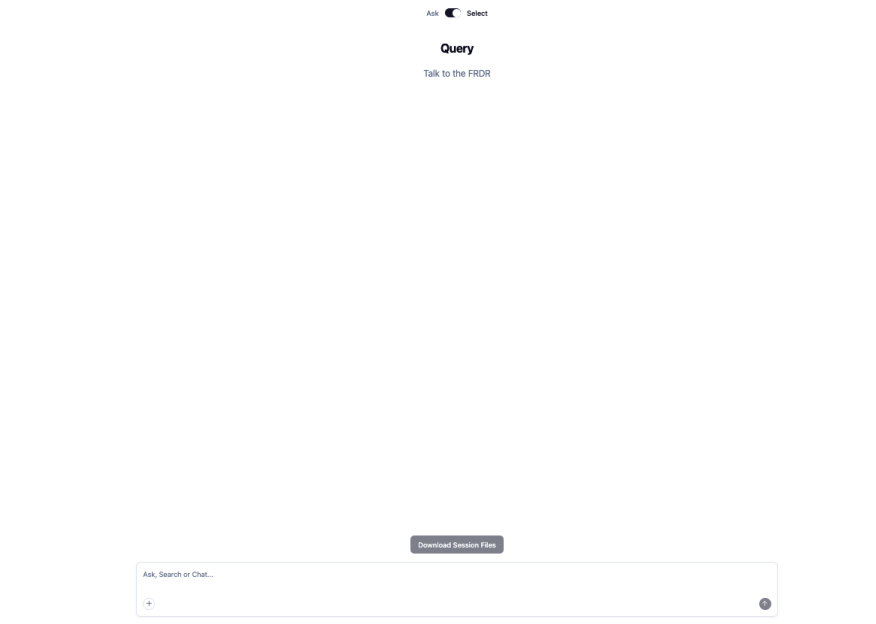

Figure 2: Query: use Select when you need tables and downloads.

8.4 Workflow D: See how much data exists and list sessions

1. Open **Inventory** (*Data Inventory*).
2. Quick start: press **Show full inventory** or **Unoperated rats only**, *or* expand filters on the left (drugs, file types, brain and apparatus fields, session type).
3. Press **Apply**. The main area shows session count, counts by file type, and a bar chart.
4. To see rows: press **View sessions (up to 500)**. A table appears when loading finishes.
5. To download files for listed sessions: use **Download Session Files** in the left panel (only after step 4 produced a non-empty table). Pick formats and confirm as in Workflow C.
6. Press **Clear** when you want to reset filters and results.

Use the small **(i)** icons on cards for short definitions (for example what counts as one *session*).

8.5 Workflow E: Inspect one session in depth

1. Open **ToolBox** (*Analysis Toolbox*).
2. In *Search or select a session...*, type part of an ID; pick a row from the dropdown.
3. Press **Load Session**. Wait until loading finishes.
4. Read **Session Info**, then **Data Points** (scroll the table). View the image if shown, and the **interactive path** diagram.
5. Read the three summary cards: distance, total checks, average check time (shown as $\log_{10}$ seconds). If distance says it is not available, press **Generate Distance?** once.

If the app says the session is not supported, try another ID or confirm with your team.

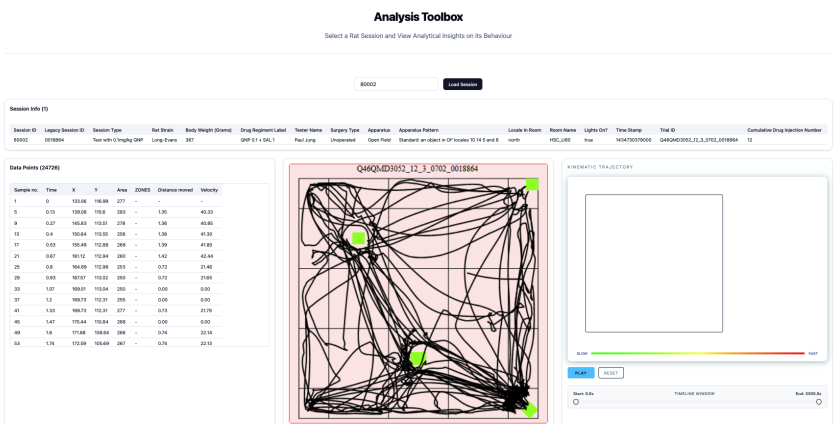

Figure 3: ToolBox: follow search → Load → tables and metrics.

8.6 Workflow F: Build a bar chart, heatmap, or line chart

1. Click **Visualizations**. You see a gallery of tiles; press **Open** on **Bar Chart** or **Heatmap**.
2. For a **bar chart**: choose **X-Axis (Group By)** and **Y-Axis**; the chart updates from your choices.
3. For a **heatmap**: choose two dimensions and a **metric** for colour.

Line chart (trends over time) is **not** listed on the gallery page. If your group uses it, ask for the direct link or bookmark it when someone shares it. On that page, choose **X-Axis (Time Period)** and a **Y-Axis** metric.

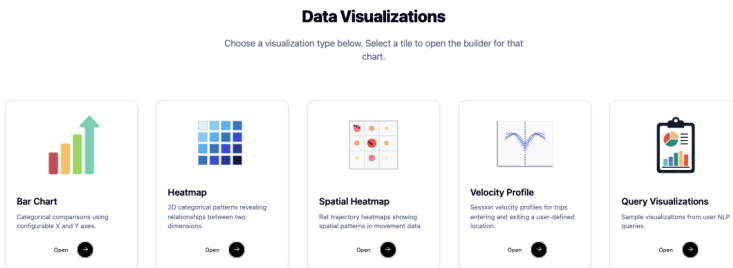

Figure 4: Visualizations: choose a tool, then Open.

8.7 Workflow G: Map where animals spent time (spatial heatmap)

1. From **Visualizations**, open **Spatial Heatmap**.
2. Under **Sessions**, tick one or more IDs; use **Clear All** to reset. Unsupported sessions are listed in a message.
3. Set **Bin size (cm)** and the **metric**: *Entries (Visits)* or *Time Spent*.
4. Optionally set **Fixed scale max** so colours stay comparable across sessions.
5. Optionally define a **home base**: enter the four corners in centimetres, press **Set Home Base**; the page can show time in that region as a percentage. Use **Clear** to remove it.
6. Move the mouse over the map to inspect bins. Use **Export Data to CSV** or **Export Map to PNG** to save your work.

8.8 Workflow H: Speed into and out of a zone (velocity profile)

1. From **Visualizations**, open **Velocity Profile**.
2. **Select session** with the search field (same idea as ToolBox).
3. **Define zone:** enter centre X, Y, and **Radius**. If you need coordinates, load the session in **ToolBox** first and read values from the data or plot.
4. Set **Max frames per side** and **Min trip frames** (short segments are treated as lingering and skipped).
5. Choose **Colouring**: *Temporal gradient* or *Time-slice highlight* (then use the slider under the chart when applicable).

6. Press **Generate**. Read the summary tiles and the chart. If nothing appears, try a larger radius or a smaller minimum trip length.

8.9 Workflow I: Compare groups on standard measures (dashboards)

1. From **Visualizations**, open **Query Visualizations**.
2. Read the title and description for the selected layout.
3. Open the **Graph Set** menu and choose a format (for example Tucci-style snapshot bars or Ballester-style lines across injections).
4. Scan the grid of panels (checking frequency, check length, recurrence, stops before returning, rest duration, and related measures depending on the set).
5. Hover points or bars for mean, error, and sample size when offered. If one panel fails, others may still load—read any warning at the top.

8.10 Workflow J: Get help or contact the team

1. Click the **info** icon (**About**).
2. Expand FAQ items for purpose, audience, data types, and downloads.
3. If you still need detail, use the button to go to **Query** and ask there, or use **GitHub** from the header for technical issues.

9 Reference

9.1 ZIP downloads (Query and Inventory)

Bulk downloads are one ZIP per job. You choose which types to include:

CSV
: Spreadsheet-friendly tables.

EWB
: Behavioural workflow files for the research pipeline.

JPG
: Still images.

MPG
: Videos.

GIF
: Short animations.

Spatial heatmap **Export to CSV/PNG** is separate from this ZIP flow.

9.2 Glossary

Ask Chat answers without a results table or Query-page ZIP download.

Select
Mode that shows a plan, SQL, and results; required for downloads on the Query page.

FRDR
Federated Research Data Repository; mentioned on Query and in download names.

Session
One behavioural trial for one animal (see Inventory tooltips).

Graph set
A named dashboard layout on **Query Visualizations**.

Home base
A rectangle on the spatial heatmap used to measure time in a region.

9.3 If something goes wrong

- **Download on Query is greyed out.** Send another message in **Select** that returns a table, and make sure that message is the **last** one in the thread.
- **Download on Inventory is greyed out.** Run **View sessions (up to 500)** and wait for a non-empty table.
- **ToolBox menu does nothing or errors.** Refresh the page; if it persists, contact your administrator (the registered address may differ by host).
- **Empty chart.** Try other axis or metric choices.
- **Stuck progress bar.** Wait for the current job to finish; refresh only if it hangs a long time.